Process Characterization Report

A-Mab Drug Substance — Ultrafiltration / Diafiltration (Step 10) — PCR-010

Process Development / Manufacturing Science & Technology (MSAT)

2026-07-24

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Field	Value
Document ID	PCR-010
Document class	Process Characterization Report
Title	Process Characterization Report: Ultrafiltration / Diafiltration (Step 10)
Product	A-Mab
Version	1.0
Effective date	2026-07-24
Status	Approved (synthetic)
Document owner	Manufacturing Science & Technology (MSAT)
Sponsor / sites	Novacyte Biologics; Novacyte Biologics — Cambridge, MA (Development) → Novacyte Biologics — Grafton, WI (Commercial DS)

Approvals

Table 1: Approval record (synthetic; signatures not applied).

Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Analytical Development	—	—	—
Reviewer	Manufacturing Science	—	—	—
Reviewer	Statistics	—	—	—

Role	Function	Name (synthetic)	Signature	Date
Approver	Quality Assurance	—	—	—

Abbreviations

A280, protein concentration by ultraviolet absorbance; AMV, analytical method validation; CPP, critical process parameter; Cpk, one-sided process capability index; CQA, critical quality attribute; DS, drug substance; DV, diavolume; HMW, high molecular weight; icIEF, imaged capillary isoelectric focusing; KPP, key process parameter; MSAT, Manufacturing Science and Technology; NOR, normal operating range; PAR, proven acceptable range; RSD, relative standard deviation; SEC-HPLC, size exclusion high performance liquid chromatography; SOP, standard operating procedure; TFF, tangential flow filtration; TMP, transmembrane pressure; UF/DF, ultrafiltration and diafiltration; WC-CPP, well-controlled critical process parameter.

Executive summary

Ultrafiltration and diafiltration is the last operation of the A-Mab drug substance process. The step concentrates the antibody to the drug substance target and exchanges the process buffer for the formulation buffer. It forms no quality attribute of the antibody, and no impurity clearance and no viral clearance are claimed for it. The outputs of the step are the drug substance concentration and the completeness of the buffer exchange. Both are measures of process performance rather than quality attributes.

This report records the characterization of the step. The study was executed under PCP-010 on a qualified scale-down model of the commercial tangential flow filtration skid, following the Stage 1 process design expectations of the lifecycle approach (U.S. Food and Drug Administration 2011) and the enhanced development approach of ICH Q8 and ICH Q11 (International Council for Harmonisation 2009, 2012). The risk assessment RA-001 carried 3 parameters into the study. Each was assessed one at a time over the range given in Table 4.1, because none of them is linked to a critical quality attribute and a multivariate design was not warranted (§2.3).

The step delivered drug substance at 75 g/L with a step yield of 97.1%, which brings the cumulative yield of the drug substance process to 83.2% (§5.1). At the set-point of 8 diavolumes the buffer exchange leaves 0.034 % of a freely permeable species in the retentate (§5.2). No effect of any parameter on product quality was detected over the ranges assessed, and all 3 parameters were classified as key process parameters (§9). No parameter of this step was classified as critical or as well-controlled critical.

The two attributes that could be altered by the step meet their acceptance criteria at commercial scale. Aggregate reaches a Cpk of 16.1 and acidic charge variants a Cpk of 4.26 over 2,000 simulated batches with every parameter inside its normal operating range (§8). The proven acceptable ranges of §7 are bounded by the ranges studied and not by a fitted model, because the step governs no critical quality attribute.

2 deviations were recorded during execution and both were retained. One was a transmembrane pressure excursion above the normal operating range, and the other was a final concentration below target on a single run. Neither took the step outside the range over which it was characterized (§13). The outcome of this step rolls up into PCMR-001.

1 Introduction

1.1 Product and unit operation

A-Mab is a humanized IgG1 monoclonal antibody produced by fed-batch cell culture at a commercial scale of 15,000 L and purified on the Novacyte platform train. The train runs from harvest and clarification through Protein A capture, low pH viral inactivation, cation exchange, anion exchange and small virus retentive filtration to ultrafiltration and diafiltration, which is Step 10 and the subject of this report. The step receives the filtrate of the small virus retentive filtration step and delivers formulated drug substance.

The step is operated as tangential flow filtration on a membrane whose pores pass buffer salts and small solutes and retain the antibody. Ultrafiltration removes permeate and concentrates the retentate. Diafiltration then adds formulation buffer at the rate at which permeate leaves, so the retentate volume is held constant while the process buffer is displaced. A final concentration adjustment brings the retentate to the drug substance target of 75 g/L. The whole operation is described in SOP-2013.

The step forms no quality attribute of the antibody. It removes no impurity that the control strategy credits, and no viral clearance is claimed for it. What the step can do to the product is mechanical. The antibody accumulates in the polarisation layer at the membrane wall, where it is more concentrated than in the bulk retentate, and it passes repeatedly through the recirculation pump. Both are routes to aggregate. Aggregate is therefore the attribute this characterization has to bound, and it is treated in §2.2 and §5.3.

1.2 Objectives

The objectives of the study were as follows.

(i) Quantify the relationship between each process parameter and the outputs of the step.
(ii) Establish the region over which the step delivers drug substance at the target concentration with the buffer exchange complete.
(iii) Establish a proven acceptable range for each parameter and confirm that the normal operating range lies inside it.
(iv) Classify each parameter against the criteria of SOP-4001.
(v) Justify the ranges to be run in Stage 2 and state this step's contribution to the control strategy.

1.3 Regulatory and scientific basis

The study follows the lifecycle approach to process validation, in which characterization belongs to Stage 1 and establishes the process understanding that the commercial control strategy rests on (U.S. Food and Drug Administration 2011). The enhanced development approach of ICH Q8 and ICH Q11 sets the framework for linking process parameters to quality attributes and for defining the ranges that follow (International Council for Harmonisation 2009, 2012). Criticality is treated as a continuum rather than as a binary label, in the sense of ICH Q9 (International Council for Harmonisation 2023b). The process model, the attribute framework and the risk methodology follow the A-Mab case study (CMC Biotech Working Group 2009).

Viral safety is established at the low pH inactivation, anion exchange and small virus retentive filtration steps, and the claims for those steps are made under ICH Q5A (International Council for Harmonisation 2023a). No viral clearance is claimed for ultrafiltration and diafiltration, so no spiking study was performed here and the modular clearance framework is not applied to this step.

1.4 Related documents

Table 1.1 lists the documents this report depends on and the document it feeds.

Table 1.1: Related documents.

Document ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCP-010	Process Characterization Plan (Ultrafiltration / Diafiltration (Step 10))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into

2 Prior knowledge and quality risk basis

2.1 Platform and prior-product knowledge

Ultrafiltration and diafiltration is a platform operation at Novacyte. The same membrane chemistry, the same channel geometry and the same buffer exchange strategy have been used for three previous humanized IgG1 antibodies (X-Mab, Y-Mab and Z-Mab) and through A-Mab clinical supply (CMC Biotech Working Group 2009). The platform history covers the mechanism this step works by, which is size based retention of the antibody at a membrane that passes salts. Because the mechanism is the same for A-Mab and the operating conditions are unchanged, prior knowledge applies directly, and this study confirms and bounds the platform ranges rather than searching for them.

Prior knowledge also sets what each parameter is expected to do. The number of diavolumes determines how completely the process buffer is exchanged. Each diavolume removes a fixed fraction of the permeable species that remain, so residual process buffer components fall exponentially with the diavolume count, and the count is set so that they fall below what the formulation tolerates. Transmembrane pressure drives the permeate flux. Beyond the point at which the polarisation layer at the membrane wall limits the flux, further pressure compresses that layer without raising throughput, and it raises the concentration and the shear the antibody sees at the wall. The final drug substance concentration sets the viscosity of the retentate and the concentration in the polarisation layer, and with them the pressure drop and the aggregation risk near the end of the concentration. It is fixed by the formulation and is reached by mass balance.

2.2 Quality attributes in scope

The step sets no critical quality attribute. Two attributes are nevertheless measured across it, and they are listed in Table 2.1 with their acceptance criteria and their criticality.

Table 2.1: Quality attributes measured across the step. Criticality and the Tool #1 score are those of the drug substance register.

CQA	Category	Acceptance	Criticality	Tool #1
Aggregates (HMW)	size variant	0–5 % HMW (SEC)	H	60

CQA	Category	Acceptance	Criticality	Tool #1
Acidic charge variants (deamidation)	charge variant	18–40 % (CEX/icIEF)	VL	4

Aggregate is the attribute at risk. The antibody is held at its highest concentration of the whole process in this step, and it is sheared at the pump and at the membrane wall. Aggregate carries high criticality and it is not reduced by any operation downstream of cation exchange, so any aggregate formed here reaches the drug substance. Acidic charge variants are measured because deamidation depends on pH and on time, and the step changes the buffer the antibody sits in. That attribute carries very low criticality, and the measurement is confirmatory rather than limiting.

Protein concentration is not a quality attribute of the antibody. It is a drug substance specification that this step is responsible for meeting, and it is reported here as a process performance outcome.

2.3 Risk-based prioritization and parameter selection

RA-001 ranked the parameters of every step by their potential impact on a quality attribute and by their potential to interact with other parameters, and it converted the ranking into a study type. No parameter of this step reached the score at which a multivariate design is required, because none of them acts on an attribute the step forms or clears. The assessment carried 3 parameters into univariate characterization and assigned no parameter of this step to a designed experiment. The three parameters are the number of diavolumes, the transmembrane pressure and the final drug substance concentration.

The ranges to be studied were set from the same assessment. For each parameter the range extends beyond the normal operating range on the side where the risk lies. It extends above the normal operating range for transmembrane pressure and for the diavolume count, and on both sides for the final concentration. The consequences of that choice for the operating ranges are taken up in §4.1 and §6.

3 Materials and methods

3.1 Scale-down model and its qualification

The studies were run on a scale-down tangential flow filtration system, EQ-TFF-142, qualified as a model of the commercial skid under SOP-1001. The model uses the same membrane chemistry, the same nominal molecular weight cut-off and the same channel geometry as the commercial membrane. It is operated at commercial equivalents for the quantities that govern the step. Membrane loading is matched in grams of antibody per square metre of membrane, crossflow is matched per unit channel cross-section, transmembrane pressure is matched directly, and the diavolume count is matched by definition because it is dimensionless. The step is operated under SOP-2013 at both scales.

The model was qualified by comparing the inputs and the outputs of the step at the two scales. The comparison covers the permeate flux at the set-point, the step yield, the final concentration and the product quality attributes of Table 2.1. EQ-TFF-142 was inside its calibration interval when the studies were executed, and its next calibration was due 2026-04-22.

One feature of the step does not scale. The recirculation volume of a small system is larger relative to its batch volume than that of the commercial skid, so a molecule passes the pump a different number of times at the two scales. The scale-down model therefore reproduces the concentration and the pressure the antibody experiences, but not the pump passage count. The consequence for the aggregate result is stated in §5.3 and again in §11.

3.2 Operation

Each run started from a representative filtrate of the small virus retentive filtration step. The retentate was concentrated at the transmembrane pressure under test until the antibody reached an intermediate concentration, diafiltered against formulation buffer for the number of diavolumes under test at constant retentate volume, and then concentrated to the final target. The permeate was collected and weighed. At the end of the run the system was chased with formulation buffer to recover the retentate held in the recirculation loop and the membrane housing, and the recovered material was combined with the retentate.

Buffers were prepared and released to platform specification. Buffer composition and the membrane lot are controlled inputs rather than characterization factors, so they were held at their platform values and were not varied.

3.3 Analytical methods

Table 3.1 lists the controlled procedures and the validated analytical methods used for this step. Protein concentration was measured by ultraviolet absorbance (AMV-3019), aggregate by size exclusion chromatography (AMV-3011) and charge variants by imaged capillary isoelectric focusing (AMV-3013).

Table 3.1: Controlled procedures and validated analytical methods for this step.

Reference	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP
SOP-2013	Operation of the Ultrafiltration / Diafiltration (Formulation) Step	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP
AMV-3011	Size-Variants (SEC-HPLC)	Method validation
AMV-3019	Protein Concentration by A280 (UV)	Method validation
AMV-3013	Charge Variants (icIEF)	Method validation

The concentration assay carries the study. It measures the parameter that is under test, it measures the drug substance attribute the step is responsible for, and it closes the mass balance. Its precision is 1.2 %RSD and its accuracy is 99.6 % of the nominal value, with a limit of quantitation of 0.5 g/L. A difference of a few grams per litre at the drug substance target is therefore well outside the uncertainty of the method, which matters for the deviation reported in §13.2. Method performance is summarized in Appendix B.

3.4 Sampling plan

Samples were taken from the retentate at the start of the concentration phase, at the end of the diafiltration phase and from the final drug substance. Concentration was measured at every sampling point, because it is needed for the mass balance. Aggregate and charge variants were measured on the starting material and on the final drug substance of each run, so that any change across the step is read as a difference on the same material rather than against a historical mean. The permeate was retained for each run and assayed for protein, which is how membrane integrity was confirmed.

3.5 Statistical methods

No statistical model was fitted for this step. A model requires a designed experiment, and the risk assessment assigned no parameter of this step to one (§2.3). The parameters were assessed one at a time, and the results are reported as measured values against the range studied. The consequence is stated plainly in §4.3. A univariate assessment cannot resolve an interaction between two parameters, and none is claimed here.

Commercial-scale capability was estimated for the drug substance attributes by Monte-Carlo simulation of 2,000 batches, with every process parameter of every step sampled inside its normal operating range. Capability is reported as a one-sided Cpk against the acceptance criterion that binds, which is the upper criterion for both attributes in Table 2.1. The simulation is a whole-process calculation and is not a property of this step alone, and §8 says what part of it this step is responsible for.

4 Study design

4.1 Factors, ranges and the knowledge space

Table 4.1 gives the three parameters with their set-points, their normal operating ranges, the ranges over which they were characterized and the classification that follows from the results of §5.

Table 4.1: Parameters, ranges and final classification. The characterization range is the range studied, not a proven acceptable range.

Parameter	Unit	Set-point	NOR	Char. range	Class	Study
Number of diavolumes	DV	8	7–9	7–10	KPP	univariate
Transmembrane pressure	psi	12	10–14	10–15	KPP	univariate
Final DS concentration	g/L	75	70–80	65–85	KPP	univariate

The characterized range is wider than the normal operating range for every parameter, but not by the same amount on both sides. For the diavolume count and for transmembrane pressure the lower edge of the normal operating range sits on the lower edge of the characterized range, so there is no characterized margin below the normal operating range for either. The margin above is 1 diavolumes and 1 psi. The final concentration was characterized 5 g/L below and 5 g/L above its normal operating range.

The asymmetry follows the direction of the risk, and it is deliberate for two of the three parameters. A diavolume count above the set-point can only make the buffer exchange more complete, and the reason to bound it from above is process time rather than quality. A count below the lower edge leaves more process buffer behind, which is the outcome that matters, and the lower edge of the range studied is also the lower edge of the normal operating range. The same holds for transmembrane pressure, where the risk lies above the set-point and the characterized range extends there. An excursion below the normal operating range in either parameter therefore falls outside the characterized region, and §6 states what follows for change control.

4.2 Univariate assessment

Each parameter was assessed one at a time. The parameter under test was set at the lower edge of its characterized range, at its set-point and at the upper edge, while the other two parameters were held at their set-points. The conditions are tabulated in Appendix A. The set-point condition is common to all three series and serves as the reference against which each edge is read.

Each run was taken through the full operation of §3.2, so every run delivered material at the drug substance target concentration and was assayed for the attributes of Table 2.1. The runs at the edges of the characterized range are the runs that carry the argument, because they are the conditions the normal operating range has to be defended against.

4.3 Limitations of the design

A one factor at a time design cannot determine whether the effect of one parameter depends on the level of another. Two of the three parameters in this step act through the same physical route, which is the concentration and the shear at the membrane wall, so an interaction between transmembrane pressure and final concentration is physically plausible. The design does not resolve it, and this report makes no claim about it. What bounds the risk instead is the size of the effects that were found at the edges of the ranges, which is reported in §5.4, and the fact that the step forms no quality attribute.

The second limitation is the scale-down model. Pump passage count is not matched between scales (§3.1), so the shear history of the small scale runs is not the shear history of a commercial batch. This is treated in §5.3 and §11.

5 Results

5.1 Concentration and mass balance

The step delivered drug substance at 75 g/L, which is the target of Table 4.1. Table 5.1 gives the mass balance of the step and of the step before it.

Table 5.1: Product mass and yield across the last two steps of the drug substance process.

Step	Unit operation	Step yield	Cumulative yield	Product mass (g)
9	Small Virus Retentive Filtration	0.984	0.857	56,232
10	Ultrafiltration / Diafiltration (formulation)	0.971	0.832	54,588

The step recovered 97.1% of the antibody presented to it. The loss is 1,644 g of product mass per batch, and it is accounted for by three routes. Retentate is held in the recirculation loop, the membrane housing and the tubing at the end of the operation, and the buffer chase of §3.2 recovers most but not all of it. A small amount of antibody adsorbs to the membrane. A small amount passes the membrane, which the permeate assay of §3.4 quantifies. None of these routes alters the quality of the material that is recovered, and the loss is a yield outcome rather than a quality outcome.

Figure 5.1 puts that yield in the context of the whole drug substance process. The cumulative yield reaches 83.2% at the outlet of this step, which is the drug substance yield of the process, and this step loses 2.9 % of the antibody entering it. The step therefore delivers 54.6 kg of drug substance per batch at commercial scale.

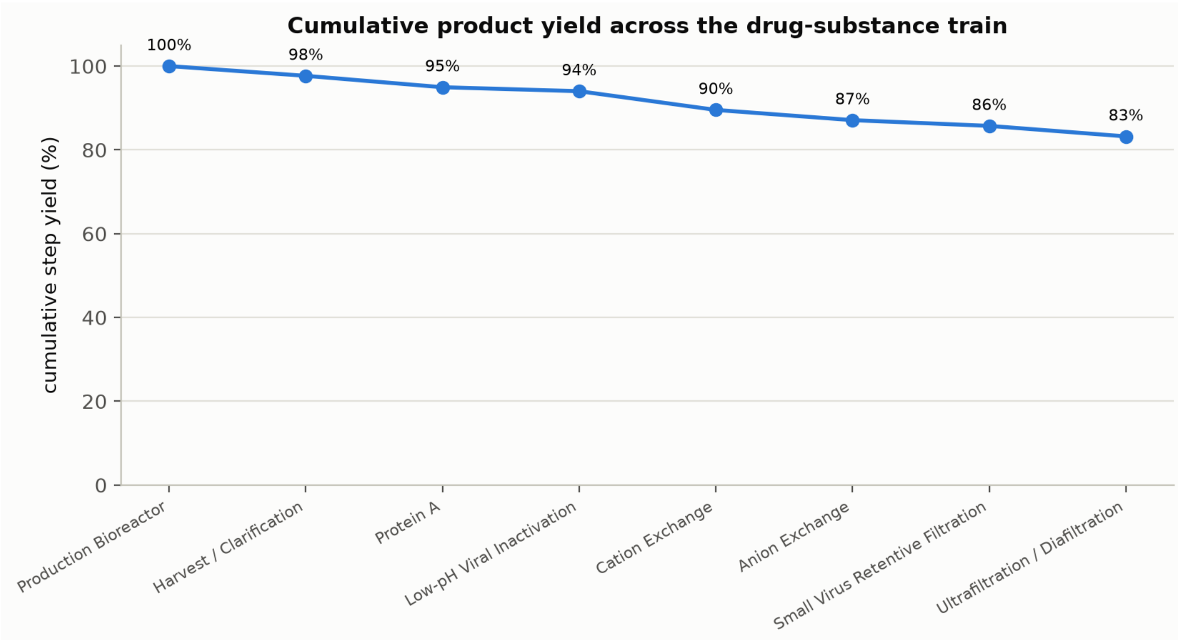

Figure 5.1: Cumulative product yield across the drug substance train. The last point is the outlet of the ultrafiltration and diafiltration step, and it is the drug substance yield of the process.

5.2 Buffer exchange

The completeness of the buffer exchange follows from the diavolume count. A species that passes the membrane freely is reduced by a fixed fraction with each diavolume of buffer added at constant volume, so its residual concentration falls exponentially with the count. At the set-point of 8 diavolumes the relation leaves 0.034 % of the starting concentration of such a species. At the lower edge of the characterized range, 7 diavolumes, it leaves 0.091 %, and at the upper edge, 10 diavolumes, 0.005 %.

Two bounds apply to that calculation. It assumes the species passes the membrane freely and that the retentate volume is held constant during the diafiltration phase, and a species that is partly retained is removed more slowly than the relation predicts. It also assumes the exchange is well mixed, which the recirculation of the retentate provides at both scales. The relation is used here to show why the step is not sensitive to the diavolume count within the range studied. Moving from the lower edge to the upper edge of the range changes the residual process buffer component by a factor of 20, and both values are far below what the formulation tolerates.

5.3 Product quality across the step

Aggregate is reported at drug substance rather than as a step increment, because the increment this step could produce is smaller than the assay can resolve on a single pair of samples. The drug substance level over the simulated commercial batches is 0.753 % HMW against an upper acceptance criterion of 5 % HMW. The pool leaving cation exchange, which is the last operation that reduces aggregate, carries 0.732 % HMW. The difference is 0.0214 % HMW, which is 0.24 of the batch to batch standard deviation of the drug substance figure.

That difference is not resolvable, and it is spread across three operations rather than this one. Anion exchange, small virus retentive filtration and this step all lie between the cation exchange pool and the drug substance, so no aggregate increment can be assigned to any one of them from this comparison. What the comparison supports is the weaker and sufficient statement. Aggregate does not grow measurably over the last three operations of the process, and this step is one of them.

Acidic charge variants behave in the same way. The drug substance level is 22.3 % against an upper acceptance criterion of 40 %, and the attribute is formed at the production bioreactor and carries very low criticality (Table 2.1). The buffer exchange changes the pH the antibody sits in, and deamidation rate depends on pH and on time. The measurement was made for that reason, and it found no change that the method could distinguish from run to run variability.

One bound applies to both attributes. The shear history of a small scale run is not the shear history of a commercial batch, because pump passage count is not matched between scales (§3.1). The aggregate result is therefore read as evidence that the mechanism does not act strongly at the concentrations and pressures studied, and not as a prediction of the commercial shear exposure. Aggregate is confirmed on every batch at drug substance release, which is where the commercial evidence comes from (§10).

5.4 Response to the parameters over their characterized ranges

No parameter changed a quality attribute over the range it was assessed in. The observations that support the classification of §9 are as follows.

The diavolume count acts on the buffer exchange and on nothing else. Its effect on the residual process buffer is large and is given in §5.2. Its effect on aggregate and on charge variants was not resolvable at either edge of the range, which is expected, because the diafiltration phase runs at constant volume and at constant concentration and adds no mechanism by which the antibody could be altered.

Transmembrane pressure acts on the permeate flux and, through the polarisation layer, on the concentration and the shear at the membrane wall. Raising it to 15 psi, the upper edge of the characterized range, raised the flux less than the pressure, which is the behaviour expected once the polarisation layer limits the flux. Aggregate at the upper edge was not distinguishable from aggregate at the set-point. The step is operated below the pressure at which the flux stops responding, which is what keeps the wall concentration close to the bulk concentration.

The final drug substance concentration acts on the viscosity of the retentate and on the concentration in the polarisation layer. At 85 g/L, the upper edge of the characterized range, the retentate is more viscous and the pressure drop along the channel is higher than at the set-point. Aggregate at that condition was not distinguishable from aggregate at the set-point, and the material met the drug substance acceptance criterion for aggregate. At 65 g/L, the lower edge, the step delivers material below the formulated target, which is a formulation failure rather than a quality failure.

These are univariate observations. They bound each parameter at the edges of its own range with the other two at their set-points, and they do not bound the corner at which transmembrane pressure and final concentration are both high (§4.3).

6 Operating ranges in place of a design space

This step has no design space. A design space is the multivariate region over which the critical quality attributes governed by a step remain within their limits, and it is defined from a model that predicts those attributes from the parameters (International Council for Harmonisation 2009). This step governs no critical quality attribute and no such model was fitted (§3.5), so the region that governs its operation is a set of univariate ranges rather than a multivariate region. The step is described here by its operating ranges and by the proven acceptable ranges of §7, and no claim of a design space is made for it.

The ranges are nested in the usual way. The normal operating range of Table 4.1 is the range the batch record allows. It sits inside the characterized range, which is the range over which the step was studied and which is the limit of what the data support. Movement inside the normal operating range is routine operation. Movement outside it is a change and is handled by the quality system, and the assessment of such a change should note that for the diavolume count and for transmembrane pressure there is no characterized margin below the normal operating range (§4.1). An excursion downward in either parameter is outside the region this study covers, and it requires its own assessment rather than a reference to this report.

7 Proven acceptable ranges

A proven acceptable range is a parameter range proven by supporting data to deliver acceptable output. For a step that governs a critical quality attribute, the range is back-calculated from the attribute limit through the fitted model, and it is reported per attribute and per parameter. Neither condition holds here. This step governs no critical quality attribute, and no model was fitted, so no proven acceptable range can be back-calculated.

The proven acceptable range of each parameter is therefore the range over which it was assessed and over which the step delivered drug substance at target concentration with the buffer exchange complete and with no resolvable change in the attributes of Table 2.1. Those ranges are the characterization ranges in Table 4.1, which are 7 to 10 diavolumes, 10 to 15 psi, and 65 to 85 g/L. The conditions that were run are in Appendix A.

Three bounds apply to these ranges. Each was demonstrated with the other two parameters at their set-points, so a combination of two parameters at their edges is not covered (§4.3). The lower edge of the range for the final drug substance concentration is an acceptable condition for product quality but not for the formulation, because material at 65 g/L is below the drug substance target. The ranges are not extrapolated, so the claim stops at the edges stated above.

8 Process capability and robustness

Capability at commercial scale was estimated for the drug substance attributes as described in §3.5. Table 8.1 gives the result for the two attributes measured across this step.

Table 8.1: Commercial-scale capability for the attributes measured across this step, from 2,000 simulated batches with every parameter inside its normal operating range.

CQA	Crit.	Spec	Acceptance	Mean ± SD	Cpk
Aggregates (HMW)	H	upper	0–5	0.753 ± 0.088	16.1
Acidic charge variants (deamidation)	VL	upper	18–40	22.3 ± 1.4	4.26

Both attributes sit far from the criterion that binds them. Aggregate reaches a Cpk of 16.1, and the mean of 0.753 % HMW is 48 standard deviations below the upper acceptance criterion of 5 % HMW. Acidic charge variants reach a Cpk of 4.26, with a mean of 22.3 % against an upper criterion of 40 %. The margin is wide for both, and the tighter of the two is the charge variant attribute rather than aggregate.

Two bounds apply to these figures. They are whole-process figures, in which every step contributes and every parameter varies inside its normal operating range, so neither is a property of this step alone. They are also estimated from qualified scale-down models rather than measured at commercial scale, and confirmation belongs to Stage 2. The attribute with the tightest capability in the drug substance is Viral clearance of MVM (parvovirus), at a Cpk of 1.51, and it is set by another step and reported in PCR-008. This step contributes nothing to it, and the consolidated position across all attributes is in PCMR-001.

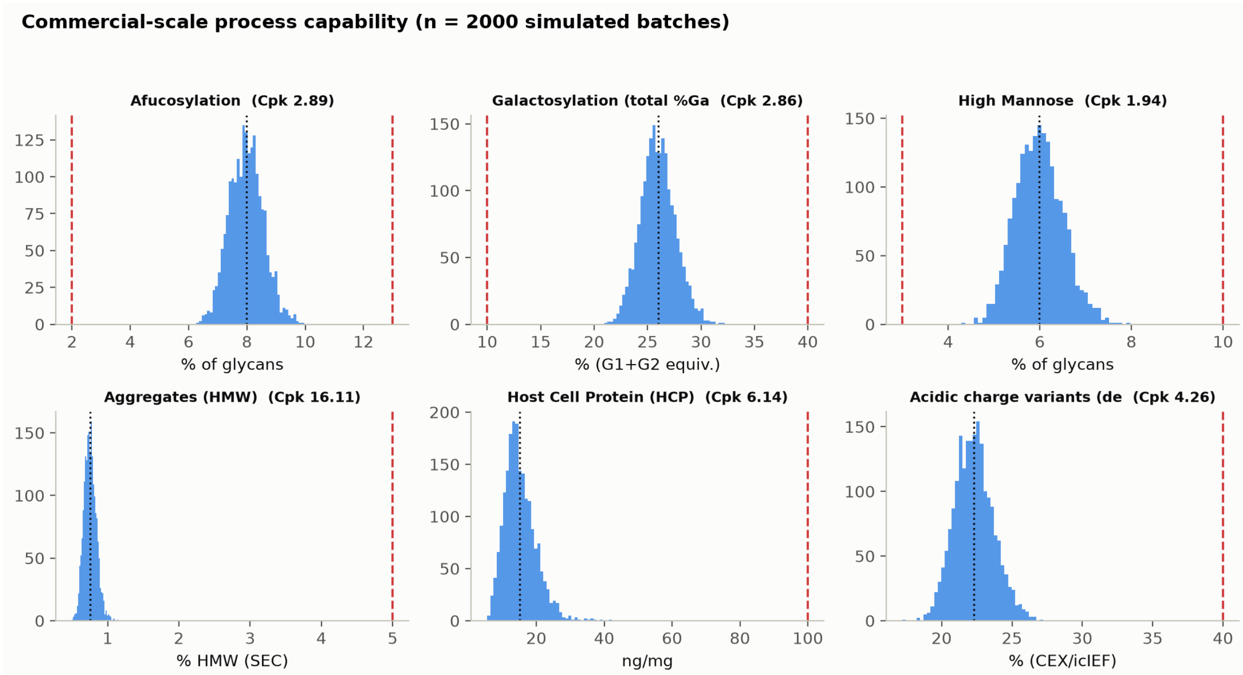

Figure 8.1: Commercial-scale capability for the drug substance attributes, from the simulated batches described in §3.5. The two panels that bear on this step are Aggregates (HMW) and Acidic charge variants. The dashed lines are the acceptance criteria and the dotted line is the mean.

Robustness is shown by the deviations as much as by the designed conditions. Both deviations of §13 moved a parameter away from its set-point during execution. In each case the excursion was detected in process, the drug substance met its acceptance criteria, and the run was retained.

9 Parameter classification

The parameters were classified against SOP-4001 from the results of §5 and from the risk of operating outside the range that supports them. A parameter is critical or well-controlled critical when its variability affects a critical quality attribute. None of these parameters does. The 3 parameters of the step were therefore classified as key process parameters (KPP), which is the class for a parameter that is controlled to keep process performance consistent rather than to control an attribute (CMC Biotech Working Group 2009). The classification is recorded in Table 4.1.

- Number of diavolumes. Classified as a key process parameter. It determines the completeness of the buffer exchange and therefore the composition of the formulated drug substance, and it had no resolvable effect on aggregate or on charge variants over the range studied (§5.4). It is set by a count of buffer volumes and is readily controlled, and the lower edge of its normal operating range is the lower edge of the range studied (§4.1), which is why it is monitored in process rather than left to the batch record alone.
- Transmembrane pressure. Classified as a key process parameter. It governs the permeate flux and therefore the duration of the step, and it had no resolvable effect on product quality up to 15 psi (§5.4). It is the parameter most likely to drift during a run, because a fouling membrane raises it at constant flux, which is what the deviation of §13.1 shows. It is controlled by the skid and trended in process.
- Final drug substance concentration. Classified as a key process parameter. It is a drug substance specification the step must meet, and it had no resolvable effect on aggregate up to 85 g/L (§5.4). It is reached by mass balance and confirmed by assay before the batch is dispositioned, so a shortfall is detected and corrected within the step rather than carried forward, as §13.2 describes.

No parameter of this step was classified as critical or as well-controlled critical, and none was classified as a general process parameter. The step therefore adds no entry to the list of quality-linked parameters that the control strategy of PCMR-001 consolidates.

10 Contribution to the control strategy

The step contributes to the control strategy in four ways.

The first is the parameter ranges. The normal operating ranges of Table 4.1 are carried into the batch record, and the characterized ranges of §7 bound what may be justified by reference to this report. The second is in-process control. Transmembrane pressure is monitored and trended through the concentration phase, which is how the excursion of §13.1 was detected, and the final concentration is confirmed by assay at the end of the step, which is how the shortfall of §13.2 was detected. The third is release testing. Aggregate and charge variants are measured on the drug substance by the methods of Table 3.1, and those results are the commercial-scale evidence for the attributes this step could affect.

The fourth follows from a deviation. The material held in the recirculation loop and the membrane housing is recovered by a buffer chase, and the volume of that chase is now specified against the hold-up volume of the system rather than as a fixed volume (§13.2). That control belongs to the equipment and transfers with it.

Two things this step does not control should be stated. It contributes nothing to viral clearance, which is established at the low pH inactivation, anion exchange and small virus retentive filtration steps and consolidated in PCMR-001. It also reduces no impurity, so the host cell protein, DNA and leached Protein A levels of the drug substance are those delivered by the chromatography steps.

11 Discussion

The step is now characterized over a range wider than it will be operated in, and the result is that it does what the platform history predicts. Its outputs are a concentration and a completed buffer exchange, and neither of the two attributes that could plausibly be altered by it changed measurably over the ranges studied. The mechanism by which this step could damage the product is mechanical, through shear at the pump and through the raised concentration in the polarisation layer at the membrane wall, and the results of §5.3 and §5.4 indicate that the mechanism does not act strongly at the concentrations and pressures this process uses.

Confidence in the scale-down model is high for the quantities that govern the step and lower for one of them. Membrane chemistry, geometry, loading, crossflow and transmembrane pressure are matched at the two scales, and the diavolume count is dimensionless. Pump passage count is not matched, and it is the variable that carries the shear history. For this reason the aggregate result of §5.3 is treated as a bound rather than as a prediction, and drug substance release testing carries the commercial evidence.

Three further limitations are worth stating. The design is univariate, so the corner at which transmembrane pressure and final concentration are both high is not covered by data (§4.3). Membrane life-cycle effects were not studied here, because the studies were run on membranes within their qualified use, and a change in membrane re-use is a change to the step. Finally, the ranges of §7 are not extrapolated, and an excursion below the normal operating range in the diavolume count or in transmembrane pressure falls outside the characterized region (§6).

The two deviations of §13 were both retained. Neither took a parameter outside the range over which the step was characterized, and neither changed a conclusion of this report. They are also useful evidence in themselves, because each one shows an in-process control detecting an excursion and a documented recovery from it.

12 Conclusions

The ultrafiltration and diafiltration step was characterized on a qualified scale-down model over ranges wider than the normal operating ranges, and it delivered drug substance at 75 g/L with a step yield of 97.1%. No parameter had a resolvable effect on a quality attribute over the range it was assessed in, and all 3 parameters were classified as key process parameters. The proven acceptable ranges of §7 are the ranges studied, and they bound the claim.

The two attributes measured across the step meet their acceptance criteria at commercial scale, at a Cpk of 16.1 for aggregate and 4.26 for acidic charge variants. Both figures are whole-process estimates from qualified scale-down models and are confirmed at commercial scale in Stage 2. 2 deviations were recorded and both were retained, and neither took the step outside its characterized region. The step is ready for Stage 2 at the ranges in Table 4.1, and its outcome rolls up into PCMR-001.

13 Deviations from the plan

2 deviations were recorded during execution of PCP-010. Both were retained, which is to say that the affected data were kept and used. Table 13.1 is the register, and each deviation is treated in turn below.

Table 13.1: Deviation register for this report.

Deviation	Summary	Detected during	Disposition
DEV-010-01	Transmembrane-pressure excursion above NOR during concentration	in-process TMP monitoring	retained
DEV-010-02	Final DS concentration below target on one run; topped up	final concentration check	retained

13.1 DEV-010-01, transmembrane pressure excursion during concentration

Transmembrane pressure reached 14.6 psi during the concentration phase of one run, which is 0.6 psi above the upper edge of the normal operating range. The excursion was detected by the in-process transmembrane pressure trend and the run was completed as written.

The investigation examined the skid, the membrane and the operating record. EQ-TFF-142 was inside its calibration interval at execution (§3.1), so a pressure measurement fault was excluded. The pressure rose during the second half of the concentration phase and returned to its set-point after the buffer flush at the end of the run, with no operator intervention and no change to the crossflow. A fixed restriction in the flow path would not recover in that way. A fouling layer that builds under the rising wall concentration of the concentration phase and is removed by the flush does. The root cause was assigned to transient membrane fouling.

The impact on the study rests on two facts. The maximum pressure stayed inside the characterized range for the parameter, whose upper edge is 15 psi, with 0.4 psi to spare. The run was therefore executed inside the region over which the step was characterized. Transmembrane pressure showed no resolvable effect on aggregate or on charge variants across that range (§5.4), and the run delivered drug substance meeting the acceptance criteria of Table 2.1. The deviation was retained and the run was kept in the assessment. Its wider value is that it shows the in-process trend detecting a drift of this size, which is why transmembrane pressure is trended in the control strategy (§10).

13.2 DEV-010-02, final concentration below target

The final drug substance concentration of one run was 71.5 g/L at the first concentration check, which is 3.5 g/L below the target of 75 g/L. The value was inside the normal operating range, 1.5 g/L above its lower edge, so the run was not outside its range. It was below the concentration the formulation requires. The shortfall was measured by ultraviolet absorbance (AMV-3019), whose precision is 1.2 %RSD (§3.3), so a deficit of this size is far outside the uncertainty of the method.

The investigation closed the mass balance on the run. The antibody that was missing from the retentate was recovered in the buffer chase and in the post-run flush, which located it in the recirculation loop and the membrane housing rather than in the permeate or on the membrane. The permeate assay of §3.4 confirmed that the antibody had not passed the membrane. The chase volume used for the run was the volume specified for the commercial skid, and the scale-down system carries a larger hold-up volume relative to its batch volume (§3.1). The root cause was assigned to residual hold-up volume.

The retentate was concentrated further until it reached the target, which required the retentate volume to be reduced by a further 4.7 % on the mass balance, and the run was then completed. That correction added a period of recirculation and therefore additional shear exposure, which is the mechanism by which this step could raise aggregate (§2.2). The additional exposure was small against the concentration phase it followed, and the drug substance from the run met the acceptance criterion for aggregate. The deviation was retained on that basis. The chase volume is now specified against the hold-up volume of the system in use rather than as a fixed volume, which is the control recorded in §10.

14 Appendices

14.1 Appendix A. Univariate assessment conditions

Table 14.1 gives the levels at which each parameter was assessed. The parameter under test was set at the level shown while the other two were held at their set-points, so the set-point row of each series is the common reference condition described in §4.2.

Table 14.1: Levels assessed for each parameter, with the normal operating range for comparison.

Parameter	Unit	Low level	Set-point	High level	NOR
Number of diavolumes	DV	7	8	10	7–9
Transmembrane pressure	psi	10	12	15	10–14
Final DS concentration	g/L	65	75	85	70–80

14.2 Appendix B. Analytical methods summary

Table 14.2 lists the validated methods used in the study, the attribute each one measures and the unit each one reports. Method performance for the concentration assay is given in §3.3.

Table 14.2: Validated analytical methods used for this step.

Method validation	Method	Attribute measured	Reportable unit
AMV-3011	Size-Variants (SEC-HPLC)	Aggregate (high molecular weight species)	% HMW
AMV-3019	Protein Concentration by A280 (UV)	Protein concentration	g/L
AMV-3013	Charge Variants (icIEF)	Acidic charge variants	% acidic

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